Forecasting S&P 500 Realized Volatility Using Machine Learning

Comparing HAR, elastic net and random forest on nested feature sets

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Abstract

We forecast the realized variance of the S&P 500 at horizons of 1, 5 and 22 trading days.

Models and benchmarks. Three models are compared: the heterogeneous autoregressive (HAR) model of Corsi (2009), the elastic net and the random forest. They are judged against two simple benchmarks: a naive Persistence forecast (the latest realized variance) and the VIX-implied variance-swap strike.

Data. The main target is Risk Lab SPY realized variance from 5-minute subsampled trades (January 2005 – October 2026). VOLARE E-mini futures (ES) measures from 2010 supply a second, full-session target and a block of jump, semivariance and quarticity features. The machine-learning models run on four nested feature sets. Every input is tested for unit roots and long memory. Non-stationary inputs are fractionally differentiated with the smallest order that passes an ADF test, chosen inside each training window.

Evaluation. An annual expanding-window walk-forward re-fits every model each year and tunes on the last two training years. Losses are QLIKE (primary) and MSE, tested with Diebold–Mariano tests and the Model Confidence Set, plus a variance-swap check.

Status. This report documents the full implementation and its **development-period** results: pseudo-out-of-sample 2010–2017, with all models together on 2015–2017. The 2018–2026 test is locked and has not been run.

Findings.

- **1-day horizon:** the random forest cuts QLIKE by 26–37% against HAR for both targets, and significantly beats the Persistence floor.
- **5 and 22 days:** several models beat HAR. But none, HAR included, significantly beats naive Persistence, except the elastic net on set B for the ES target at 5 days.
- **RQ2:** once the elastic net is well behaved, the forest does not extract more from richer information than the elastic net does.
- **Horizon:** gains do not grow with the horizon.

> **Status (16 October 2026; implements proposal v5.4).** Everything in the proposal is implemented and runs end to end in MATLAB: data alignment and publication lags; unit-root and stationarity diagnostics, fractional differencing chosen inside each training window, and the long-memory estimators; the Persistence benchmark; all three models on all four feature sets, at every horizon, for both targets; the walk-forward engine and the full evaluation suite; and all robustness checks, including the new FFD-lag variant. The numbers in Section 7 are **development-period** results. The locked 2018 – October 2026 test is gated behind a configuration flag and has not been run. Section 10 lists the decisions to settle before the test is unlocked; proposal v5.4 itself leaves two of them open (missing-day treatment and ES holiday rows).

1. Introduction and Motivation

Volatility sets the price of every option, the size of risk limits and how much a fund can safely invest. It is not observed directly. It can be measured after the fact by summing squared intraday returns, which gives **realized variance (RV)** (Andersen, Bollerslev, Diebold & Labys, 2003). For day t, with M intraday returns $r_{t,i}$:

$$\mathrm{RV}_t = \sum_{i=1}^{M} r_{t,i}^2 \tag{1}$$

Five-minute sampling is the standard compromise between information and microstructure noise, and it is the frequency used by all three reference papers.

Forecasting S&P 500 realized volatility is **crucial for volatility trading with S&P 500 options, including variance swaps**. A one-month variance-swap strike is approximately VIX^2. Anyone who forecasts realized variance better than that price knows when volatility looks cheap or expensive. The same forecasts drive volatility-targeting funds, value-at-risk limits and margin models.

Two structural facts motivate a machine-learning comparison:

- **HAR is a strong but linear baseline.** Corsi's (2009) model predicts future RV from its daily, weekly and monthly averages. Three regressors are enough to approximate the long memory of volatility (Section 4.3). The reference papers disagree on whether ML beats it:
  - Kılıç (2026) finds that ML sometimes improves on HAR but does not systematically beat a broader econometric set.
  - Christensen, Siggaard & Veliyev (2023) find that ML does beat the HAR family, especially at longer horizons.
  - Rahimikia & Poon (2026) find that ML wins on most days, but HAR does better on extreme-volatility days.
- **Richer information is available, but only some of it is RV history.** Option-implied volatility, rates, macro and financial-stress indices can all enter as features. So can VOLARE E-mini futures decompositions (jumps, semivariances, quarticity), available from late 2009. Linear models struggle with many correlated inputs; trees can capture interactions and regimes. Many of these inputs are close to unit-root processes, which is why the design tests them and, where needed, fractionally differentiates them.

Our contribution is deliberately narrow:

- **Three models:** HAR, the elastic net and the random forest, all on the same horizons and walk-forward design.
- **A naive Persistence benchmark** ("tomorrow looks like today") that every forecast must beat.
- **HAR as the fixed benchmark** on its own lags (set A).
- **Four nested feature sets for the two ML models.** This lets us ask whether they beat HAR, and also whether the non-linear model extracts more than the linear one from the implied-vol, macro/stress and ES features.

How to read this report.

- Sections 2–6 follow the proposal's structure and give every definition and formula needed to reproduce the work.
- Section 7 reports the development-period results.
- Section 8 relates them to the reference papers.
- Sections 9–11 cover applications, risks and open decisions, and the implementation.
- The appendices give a feature dictionary, a line-by-line compliance check against proposal v5.4, the output files and a glossary.

2. Research Questions and Hypotheses

Research question	Hypothesis	How it is tested
RQ1. With volatility history alone (set A), do the elastic net or random forest beat HAR (and the naive Persistence benchmark) out of sample?	At $h = 1$ gains are small; they grow at $h = 5$ and $h = 22$, consistent with Christensen et al. (2023).	Pairwise DM tests and the MCS on set A (Persistence, HAR, EN, RF), by horizon.
RQ2. Does the random forest get more improvement from implied-vol, macro/stress and ES features than the elastic net?	The forest's non-linearity and interactions let it extract more from sets B–D, so its loss reduction from set A exceeds the elastic net's.	Per horizon and target, the change in QLIKE/MSE from A→B, A→C and A→D for RF vs EN. A DM test on the difference-in-differences of losses (eq. 15), plus the MCS.
RQ3. Does the full-session ES target change the economic conclusion relative to SPY intraday RV?	ES captures about 95% of close-to-close variance (SPY RV about 56%), so ES-target forecasts should align better with a variance-swap payoff.	The pipeline is repeated on VOLARE ES (2010–2026), and the variance-swap comparison is run on both targets.

The three questions are separable: a null finding on any one of them is still useful. In particular, we do not assume that implied volatility helps:

- the set A → B comparison tests whether implied-vol features add forecasting power;
- the set-B coefficients and importances (Section 7.8) show which part of the block does the work.

Success criterion (proposal Section 6). On the locked test, the design counts as successful if both of the following hold:

1. the elastic net or random forest has statistically lower QLIKE than HAR at one horizon or more, and the MCS includes the winner;
2. the variance-swap check is directionally consistent.

For RQ2 the question is answered either way, by the sign and significance of the RF-minus-EN gain. A well-documented failure to beat HAR is also a legitimate outcome (Kılıç, 2026). **Persistence is the floor:** a forecast that does not beat it is reported as uninformative at that horizon. The pipeline computes all of this automatically (Section 7.11).

3. Data

The team's cleaned inputs are read unchanged from the project's data folder; every column is described in the companion *Data Description*. No file is simply "five-minute data". Each delivers daily measures built from a stated intraday sampling frequency, or daily and weekly macro series.

File (source)	Sampling and measure	Coverage	Role in the project
Risk Lab SPY (Dacheng Xiu)	Daily annualized SPY volatility from intraday trades, regular session: rv5 (5-min subsampled), rv15, QMLE	2 Jan 1996 – 5 Oct 2026; 7,707 days	**Main target** (rv5); set-A lags; rv15 and QMLE as robustness
VOLARE ES	Daily variances from 1-second ES mid-quotes over the Globex session (18:00–17:00 ET): RV, bipower, ± semivariances, quarticity at 1-min and 5-min; realized kernel	30 Sep 2009 – 30 Sep 2026; 4,397 rows	**Second target** (rv5); set D; rv1 and kernel as robustness
Oxford-Man Realized Library (archived)	Daily S&P 500 index rv5, bipower, negative semivariance, cash session	3 Jan 2000 – 25 Feb 2022	Cross-check of the SPY target (correlation 0.968); pre-2010 jump robustness
Bloomberg IV surface	Daily SPX implied vol, 7 moneyness levels × 6 tenors (1–24 months)	3 Jan 2005 – 1 Oct 2026; 5,471 days	Set B: ATM level, skew, term slope; fixes the 2005 sample start
Cboe indices	Daily VIX, VIX9D, VIX3M, VIX6M, VIX1Y, SKEW	VIX from 1990; VIX3M from Sep 2009; VIX9D from Jan 2011	Set B (log VIX, VRP, term structure); VIX^2 is the variance-swap strike
`features_daily.csv`: market	Daily S&P 500 OHLC and returns; Hang Seng; VVIX, VXN, VXD, RVX	1990 – 6 Oct 2026; VVIX from Mar 2006	Set B; Hang Seng in set C
`features_daily.csv`: rates	Daily 10Y−3M spread and 10Y yield (FRED)	1990 – Oct 2026	Set B
`features_daily.csv`: macro and stress	EPU, CISS, real-time ADS, news sentiment, GPR (daily); NFCI, STLFSI4 (weekly)	1990 – Oct 2026; real-time ADS from Dec 2008	Set C
`features_daily.csv`: Fama–French	Daily factor returns, published about a month late	to Aug 2026	Not used (not available in real time)
NYSE calendar	NYSE sessions and early closes	1990 – 6 Oct 2026 (9,258 sessions)	Master calendar for alignment and FFD; early-close flag in set C

About the calendar. We take the NYSE calendar from the dates of `features_daily.csv`; the data team verified these sessions are identical to the NYSE calendar (special closures excluded). Proposal v5.4 names a separate calendar file (`nyse_trading_calendar_1990_2026.csv`, 9,318 sessions to end-2026) that is not in the data folder. It would only add sessions after 6 October 2026, which no forecast uses.

3.1 Sample design

Sample	Dates	Size
Main sample	January 2005 – 5 October 2026 (starts when the implied-vol surface begins)	5,440 Risk Lab days

Sample	Dates	Size
Development	**2005–2017**	3,272 days
Locked test	**2 January 2018 – 5 October 2026**	2,201 sessions; 2,168 with a Risk Lab value
ES models and set D: development	2010–2017	2,013 NYSE days
ES models and set D: locked test	Same test, through 30 September 2026	—

Pre-2005 Risk Lab data enter only a VIX-only robustness check.

3.2 Missing data, holidays and staggered starts

- **Calendar.** Every series is aligned to the NYSE trading-day calendar, and FFD (Section 4.3) is computed over NYSE sessions. Weekends and exchange holidays are therefore not missing observations. Weekly NFCI and STLFSI4 are placed on their release date and held until the next release.
- **Late starts.** A series that begins after the sample start has its features, and its FFD transform, only from its first date plus the FFD window ℓ_τ. Earlier rows are unavailable for that feature. This applies to:
    - VVIX (Mar 2006);
    - real-time ADS (Dec 2008);
    - VIX3M (Sep 2009);
    - the ES/set-D measures (Sep 2009);
    - VIX9D (Jan 2011).

  Comparisons across sets use common days only.
- **Missing trading days.** We reproduced the proposal's counts exactly from the raw files (`data_audit.m`):
    - Risk Lab SPY lacks 33 of the 5,473 NYSE sessions from 3 Jan 2005 to 5 Oct 2026, all in the test period (5–9 Feb 2018 and 28 isolated days). Development has none, and ES has no missing NYSE session.
    - Within their coverage, some inputs also lack values on NYSE days: Hang Seng 251 (Hong Kong holidays), T10Y3M and DGS10 39 (bond-market holidays), VVIX 61 (59 in 2006), RVX 7, VXD 3.
    - Targets are never filled, and ES is never a stand-in for a missing SPY target.

Horizon	Test origins	Target window has a missing day	More than 20% of target window missing	22-day lag window has a missing day
$h = 1$	2,200	33	33	531
$h = 5$	2,196	143	12	527
$h = 22$	2,179	532	19	510

Affected Risk Lab SPY test origins (reproduced from the raw files; matches proposal v5.4, Section 3.2).

- **Provisional treatment (an open decision).** Proposal v5.4 states that the treatment of missing days "will be decided now that the counts are measured". Until the team decides, the code applies four rules:
    1. a multi-day target is dropped if more than 20% of its window is missing;
    2. weekly and monthly RV lags are built from the observed days if at least 80% of the window is present;
    3. an origin whose own RV_t is missing is dropped, for every model alike;
    4. on a day when a daily input has no value (e.g. a bond holiday), its last published value is used, which uses no future information.

  All four rules are switches in `rv_config.m` / `build_design.m`. None affects the development results, because development has no missing Risk Lab days.
- **ES holiday sessions.** VOLARE has 121 rows on non-NYSE dates (abbreviated Globex holiday sessions). Their variance is rolled into the next NYSE day, so each target spans one close-to-close interval; dropping them is a robustness check. This handling is pending the professor's confirmation.

3.3 Limitations

- **VOLARE coverage.** VOLARE has no SPY or index series; ES is its only S&P 500 instrument. It is refreshed monthly, and ES ends 30 September 2026.
- **ES session timing.** The ES day is inferred as 18:00–17:00 ET, and roll dates are not flagged.
- **Missing feature types.** We lack limit-order-book and news-text features (central to Rahimikia & Poon) and the firm-level implied vol used by Christensen et al.
- **Overnight move.** SPY intraday RV omits the overnight move, so it covers only about 56% of close-to-close variance. ES captures about 95%, which is why it is the preferred target for the variance-swap check.

4. Methodology

Forecasts are produced at the close of each NYSE day t for horizons $h \in \{1,5,22\}$ trading days. Figure 1 summarizes the pipeline.

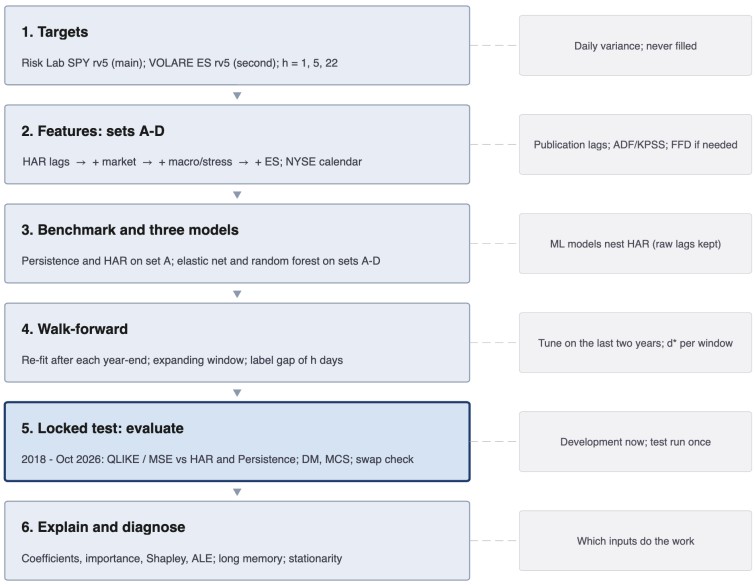


Figure 1. Forecasting pipeline. Persistence and the three models share targets, feature sets and the walk-forward engine; only the fitting step differs.

4.1 Targets

For SPY, Risk Lab reports an annualized volatility σ_t^{rv5}. We convert it to a daily variance, held in %² (i.e. decimal × 10^4), and form horizon averages:

$$\mathrm{RV}_t^{\mathrm{SPY}} = \frac{(\sigma_t^{\mathrm{rv5}})^2}{252}, \qquad \mathrm{RV}_{t+1:t+h} = \frac{1}{h}\sum_{k=1}^{h} \mathrm{RV}_{t+k} \tag{2}$$

For ES, VOLARE rv5 is already a daily variance of 5-minute mid-quote returns over the Globex session. Any linear forecast below the training-window minimum of daily RV is floored at that minimum (Christensen et al., 2023).

4.2 Nested feature sets

Set	Contents (exact features: Appendix A)	First date
A: HAR lags	Daily, weekly (5-day) and monthly (22-day) averages of the target's own RV	Jan 2005 (SPY), Jan 2010 (ES)
B: + market	A + negative-return averages (d/w/m), Parkinson range (d/w), log VIX, VRP $= \mathrm{VIX}^2/252 - \mathrm{RV}^{(m)}$, IV surface (ATM 1M/3M/1Y, 1M skew, 1Y−1M term slope), 10Y yield and 10Y−3M spread, VVIX, SKEW, VIX9D−VIX, log VXN, log RVX	Jan 2005; VVIX, VIX9D from their start + ℓ_τ

Set	Contents (exact features: Appendix A)	First date
C: + macro/stress	B + real-time ADS, log EPU, news sentiment, log GPR, CISS, NFCI, STLFSI4, squared Hang Seng return, calendar flags	Jan 2005; real-time ADS from Dec 2008
D: + ES measures	C + ES continuous/jump averages (d/w/m), ± semivariances (d/w), $\sqrt{\mathrm{RQ}} \times \mathrm{RV}$, realized kernel, overnight variance $\max(\mathrm{ES} - \mathrm{SPY}, 0)$ (d/w)	Jan 2010

How the sets are used:

- **HAR** is estimated on set A only.
- **The elastic net on set A** is a regularized HAR.
- **On sets B–D,** both ML models receive the HAR lags plus the extras, so they nest HAR.
- **Jump, semivariance and quarticity measures** enter **as features**, not as separate HAR-J / SHAR / HARQ models.

From the VOLARE bipower variation BV_t and signed semivariances, the set-D inputs are:

$$J_t = \max(\mathrm{RV}_t - \mathrm{BV}_t,\, 0), \quad C_t = \mathrm{RV}_t - J_t, \quad \mathrm{RS}_t^{+} = \sum_i r_{t,i}^2\, \mathbf{1}\{r_{t,i} > 0\}, \quad \mathrm{RS}_t^{-} = \sum_i r_{t,i}^2\, \mathbf{1}\{r_{t,i} < 0\} \tag{3}$$

4.3 Stationarity and long memory

Why it matters. Both ML models learn a mapping on the training window and apply it to later years, so their inputs must behave similarly in both. Two kinds of input cause trouble:

- VIX, rate and stress levels are close to unit-root processes;
- RV is highly persistent, with slowly decaying autocorrelations (long memory).

When an input drifts, its test values can fall outside the training range. A random forest cannot extrapolate beyond the leaf values it has seen. Elastic-net coefficients fitted to trending, correlated levels change from one re-estimation to the next.

The fix. Integer differencing restores stationarity but removes the memory that HAR exploits. **Fractional differencing** (López de Prado, 2018, ch. 5) is the middle ground. With the lag operator L ($L^k X_t = X_{t-k}$) and order $d \geq 0$:

$$(1-L)^d X_t = \sum_{k=0}^{\infty} w_k\, X_{t-k}, \qquad w_0 = 1, \quad w_k = -w_{k-1}\frac{d-k+1}{k}\ (k \geq 1) \tag{4}$$

For $d = 1$ the weights are $\{1, -1, 0, \ldots\}$ (ordinary differences). For $0 < d < 1$ they decay slowly, so part of the memory survives. The fixed-width-window version (FFD) truncates the weights once they fall below a threshold τ, so every observation is built from the same window. The order d is set to the smallest grid value that passes the ADF test at 95%:

$$\tilde{X}_t(d) = \sum_{k=0}^{\ell_\tau} w_k\, X_{t-k}, \qquad \ell_\tau = \max\{k : |w_k| \geq \tau\}, \qquad d^{*} = \min\{d \in \mathcal{G} : \mathrm{ADF}(\tilde{X}(d)) < c_{0.95}\} \tag{5}$$

Implementation.

- **Grid and threshold.** $\mathcal{G} = \{0, 0.05, \ldots, 1\}$ and $\tau = 10^{-3}$, fixed in advance. This gives windows of at most about 70 sessions.
- **The tests.** ADF uses a constant, the Schwert lag length and the MacKinnon 5% critical value. KPSS tests level stationarity with a Bartlett long-run variance and the 5% critical value 0.463. Both are coded directly, because MATLAB's `adftest` and `kpsstest` belong to the Econometrics Toolbox, which is not installed.
- **Which inputs.** Every input except the raw HAR lags and the two calendar flags is a candidate. Each candidate's d^{*} is chosen on the training window's data only, and re-chosen at every walk-forward re-fit.
- **What stays raw.** The target is never transformed, and the same raw HAR lags enter both ML models, so they still nest HAR.
- **Robustness.** An FFD variant in which the ML lag block is replaced by FFD log RV is a robustness check.

Diagnosis on the development sample (2005–2017; 2010–2017 for ES and set D; `stationarity.m`). ADF and KPSS were run on both targets and every input, in levels and, for positive series, in logs: 73 tests in all.

Verdict	Count	Series
ADF rejects a unit root **and** KPSS rejects stationarity ("persistent / long memory")	64	Both targets and the HAR lags; returns, ranges, VIX, VRP, skew, term slope, VVIX, SKEW, cross-market IV; ADS, EPU, news, GPR, NFCI, STLFSI4; every ES measure
ADF cannot reject a unit root	8	ATM 3M and 1Y implied vol, 10Y−3M spread, 10Y yield, CISS (level)
Stationary on both tests	1	$\sqrt{\mathrm{RQ}} \times \mathrm{RV}$

The dominant pattern, where ADF rejects but KPSS still rejects, is exactly what Bhatti (2026) describes for long-memory series. ADF declares stationarity while KPSS keeps rejecting under strong persistence, and neither identifies the minimal d.

Under eq. 5, only the unit-root inputs are transformed. Their d^* and retained correlation $\rho = \mathrm{corr}(\tilde{X}(d^*), X)$ are:

Input	d^* (full development sample)	ρ (memory retained)	Inside the walk-forward (set-D folds): d^* range, share of folds differenced
ATM 3M implied vol	0.10	0.992	0–0.10, 66%
ATM 1Y implied vol	0.20	0.969	0.10–0.15, 100%
10Y−3M spread	0.25	0.968	0.15–0.25, 100%
10Y yield	0.30	0.960	0.15–0.25, 100%
CISS	0.10	0.995	0–0.15, 33%
ATM 1M IV, log RVX, STLFSI4	0	1	0–0.10 in some folds, 33%

Fractional differencing therefore changes very little here. The orders are small, and the transformed inputs keep 96–99.5% correlation with the originals. What it removes is the slow drift in rates and long-dated implied vol.

Memory of RV. We estimate d in two ways:

- **Local Whittle** (Robinson, 1995), which minimizes over the first m Fourier frequencies λ_j, with periodogram $I(\lambda_j)$:

$$\hat{d} = \underset{d}{\mathrm{argmin}} \left[\ln\left(1/m \sum_{j=1}^{m} \lambda_j^{2d} I(\lambda_j) \right) - 2d/m \sum_{j=1}^{m} \ln \lambda_j \right], \qquad \mathrm{se}(\hat{d}) = 1/2\sqrt{m} \tag{6}$$

- **Bhatti's detectability-boundary estimator** d_{LTM}**,** as a cross-check. For each d on a grid the series is fractionally differenced, and $\ln|\hat{\rho}(k)|$ is regressed on $\ln k$ over lags 20 to $K_2 \leq 200$. d_{LTM} is the d with the largest regression RMSE. In his Monte Carlo the boundary sits below the true order by about $d_0(T) = 5.81\, T^{-0.367}$, and short windows bias it downward.

Series (development data)	T	Local Whittle $\hat{d}$, $m = T^{0.5}$	Local Whittle $\hat{d}$, $m = T^{0.65}$	d_{LTM}	$d_0(T)$
SPY RV 1996–2017, level	5,539	0.37 (0.06)	0.53 (0.03)	0.50	0.25
SPY RV 1996–2017, log	5,539	0.61 (0.06)	0.60 (0.03)	0.92	0.25
SPY RV 2005–2017, level	3,272	0.52 (0.07)	0.63 (0.04)	0.44	0.30
SPY RV 2005–2017, log	3,272	0.61 (0.07)	0.56 (0.04)	0.40	0.30
ES RV 2010–2017, level	2,013	0.47 (0.08)	0.40 (0.04)	0.08	0.36
ES RV 2010–2017, log	2,013	0.56 (0.08)	0.57 (0.04)	0.42	0.36

What the estimates say:

- **Local Whittle:** d is between 0.37 and 0.63 for every series. That is classic long memory, many standard errors from zero.
- d_{LTM} **is erratic:**
  - 0.40–0.44 on the 2005–2017 SPY window (about 0.7 after adding Bhatti's offset);

- 0.92 for log RV over 1996–2017;
- 0.08 for ES levels on 2,013 observations, close to his 2,000-observation minimum.

As the proposal anticipates, it is reported as a cross-check and not used to choose d.

- **Implication:** volatility memory decays over months. That is why HAR works, and why every ML set keeps the raw HAR lags.

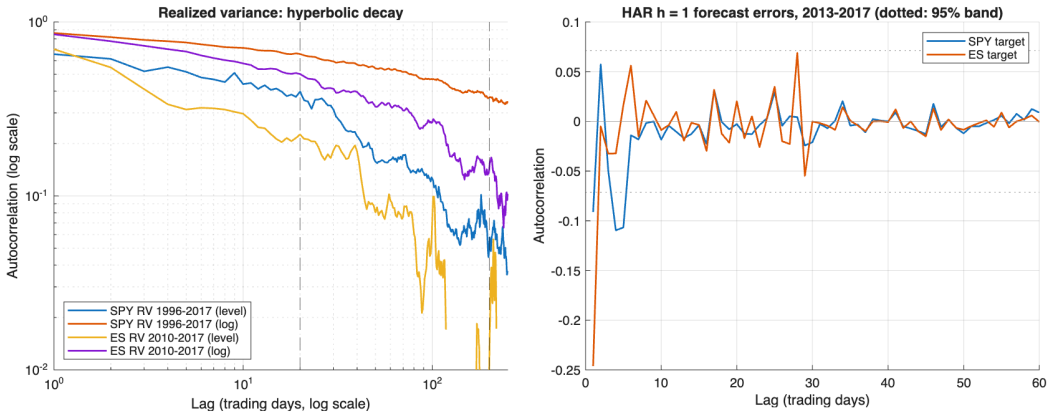


Figure 2. Left: autocorrelations of realized variance decay hyperbolically — roughly straight lines on log-log axes (dashed lines mark the 20–200 lag window of the tail regression). Right: the 1-day HAR forecast errors have almost no autocorrelation beyond the first few lags (dotted: 95% band).

4.4 Benchmarks and comparison grid

Two simple benchmarks sit beside the three models; neither is estimated.

- **Persistence** is the naive benchmark: the coming h-day average equals the latest h-day average of realized variance. For $h = 1$, tomorrow's forecast is today's variance:

$$\widehat{\mathrm{RV}}^{\mathrm{P}}_{t+1:t+h} = \frac{1}{h}\sum_{j=0}^{h-1} \mathrm{RV}_{t-j}\,, \qquad \widehat{\mathrm{RV}}^{\mathrm{P}}_{t+1} = \mathrm{RV}_t \tag{7}$$

It is the floor every forecast must clear, including HAR. - **The VIX-implied strike** VIX^2 is the market benchmark, used only in the variance-swap check.

Forecast	Type	Inputs	Estimation	Owner	Code
Persistence	Naive benchmark	Latest RV only	None	James	`fit_persistence.m`
HAR	Model 1	Set A	OLS per horizon	James	`fit_har.m`
Elastic net	Model 2	Sets A–D	Penalized regression, tuned	Aryan	`fit_en.m`
Random forest	Model 3	Sets A–D	Bagged trees, tuned	Aryan	`fit_rf.m`
VIX² strike	Market benchmark	VIX	None; variance-swap check only	All	`evaluate_phase.m`

4.5 Model 1: HAR

There is one ordinary-least-squares specification per horizon, on the variance scale (Corsi, 2009):

$$\begin{aligned} \mathrm{RV}_{t+1:t+h} &= \beta_0 + \beta_d\,\mathrm{RV}_t + \beta_w\,\mathrm{RV}^{(w)}_t + \beta_m\,\mathrm{RV}^{(m)}_t + \varepsilon_{t+h}, \\ \mathrm{RV}^{(w)}_t &= \frac{1}{5}\sum_{j=0}^{4}\mathrm{RV}_{t-j}\,, \qquad \mathrm{RV}^{(m)}_t = \frac{1}{22}\sum_{j=0}^{21}\mathrm{RV}_{t-j} \end{aligned} \tag{8}$$

It has four parameters, estimates in milliseconds, and is the literature's standard benchmark.

4.6 Model 2: Elastic net

The elastic net is a penalized linear regression on the active feature set $\mathbf{x}_t$, with target $y_{t+h} = \mathrm{RV}_{t+1:t+h}$:

$$\min_{\beta_0,\boldsymbol{\beta}} \frac{1}{2N}\sum_{t=1}^{N}(y_{t+h} - \beta_0 - \mathbf{x}_t^{\top}\boldsymbol{\beta})^2 + \lambda\left[\alpha\|\boldsymbol{\beta}\|_1 + \frac{1-\alpha}{2}\|\boldsymbol{\beta}\|_2^2\right] \tag{9}$$

- **The mixing parameter** α blends ridge ($\alpha = 0$) and lasso ($\alpha = 1$).
- **Tuning.** Both $\alpha \in \{0, 0.1, 0.25, 0.5, 0.75, 0.9, 1\}$ and a 50-point λ path are tuned on validation QLIKE, so ridge and lasso are special cases rather than separate models.
- **Implementation.** MATLAB's `lasso` handles $\alpha > 0$; the ridge case is solved in closed form in the same scaling. Predictors are standardized with training-window statistics only.
- **Reporting.** Following Kılıç (2026), the selected coefficients are reported as a result (Section 7.8).

4.7 Model 3: Random forest

The random forest averages many regression trees. Each tree is grown on a bootstrap sample, with a random subset of predictors tried at every split (MATLAB `TreeBagger`). Trees capture interactions and regimes, need little tuning and supply impurity-based importance. With B trees T_b:

$$\widehat{\mathrm{RV}}_{t+1:t+h} = \frac{1}{B}\sum_{b=1}^{B} T_b(\mathbf{x}_t) \tag{10}$$

- **Tuning.** Bayesian optimization (`bayesopt`, 12 evaluations of 100-tree forests on validation QLIKE) chooses the minimum leaf size (1–200) and the number of predictors per split (1 to p).
- **Final forest:** 300 trees.
- **A key limitation.** A forest predicts averages of training targets, so its forecasts are always positive. It also cannot forecast outside the range of volatility it has seen.

4.8 What we do not estimate

The following are out of scope:

- LogHAR, LevHAR, HAR-X, HAR-J/CJ, SHAR and HARQ;
- $\mathrm{VIX}^2/252$ as a forecasting model;
- gradient boosting, neural nets and LSTMs;
- forecast combinations.

VIX^2 remains the **market strike** in the variance-swap test, not a competing forecast. The optional exploratory boosting run on the winning feature set has not been carried out.

5. Validation Design

Days are never shuffled, and every date below is an NYSE session.

- **Boundaries.**
  - SPY: development origins 3 Jan 2005 – 29 Dec 2017. Locked-test origins run from 2 Jan 2018 to the last session whose h-day target ends by 5 Oct 2026.
  - ES and set D: development from 4 Jan 2010; test targets end by 30 Sep 2026.
- **Re-estimation dates.** After the close of the last session of year Y (date T_Y), every model is re-fitted on all data through T_Y. The window expands from 2005 for SPY and from 2010 for ES and set D. The re-fitted model is used unchanged for every origin in year $Y + 1$, so the locked test has nine re-fits, at the last sessions of 2017 through 2025. **Development pseudo-out-of-sample years use the same rule once the window holds five years: SPY 2010–2017, ES and set D 2015–2017.**

- **Tuning.** Hyperparameters are chosen by fitting on the window minus its last two years and validating on those two years. The model is then re-fitted on the whole window. d^* and standardization use the training window only.
- **Label gap.** A row with origin s is used in the re-fit at T_Y only if its target has been observed. The last h origins of each window are therefore dropped, and the same gap separates the inner fit from the two tuning years:

$$\mathcal{D}_Y(h) = \{(\mathbf{x}_s,\ \mathrm{RV}_{s+1:s+h}):\ s + h \le T_Y\} \tag{11}$$

- **Rolling window.** A rolling 10-year window (8-year for ES) is a robustness check.
- **Availability timing.** The forecast origin is the end of NYSE day t. An input dated t enters only if it is published by then; later releases are shifted:

Input	Value for day t is known	Shift (sessions)
Risk Lab SPY RV; S&P 500 OHLC; Cboe VIX family, VVIX, VXN, VXD, RVX; Bloomberg IV surface	End of day t	0
VOLARE ES measures	17:00 ET on t (Globex close)	0
Hang Seng	04:00 ET on t, before the NYSE open	0
Treasury T10Y3M, DGS10	Late afternoon of t	0
EPU	Morning of $t + 1$	1
ECB CISS	About one business day later	2
SF Fed news sentiment; classic daily GPR	Weekly updates	5
ADS	Real-time vintage released before t (current vintage lagged 5 days before Dec 2008, training rows only)	Built in
NFCI, STLFSI4	Weekly; placed on release date	Built in
Fama–French factors	About one month later	Not used

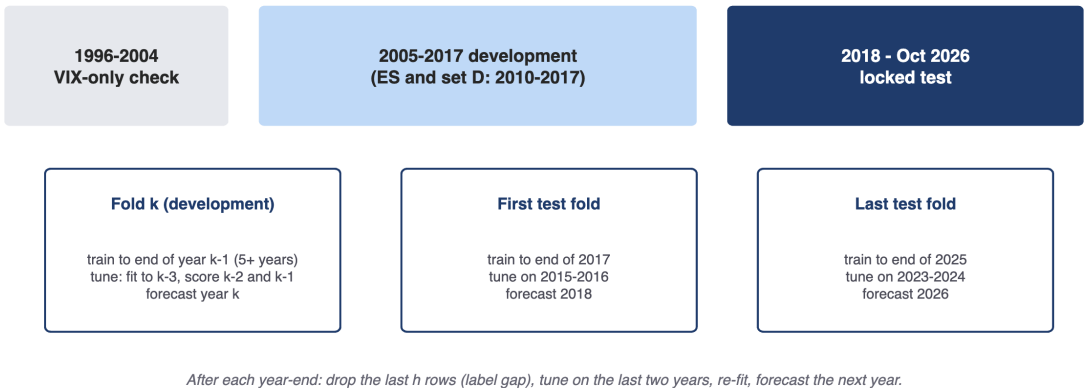


Figure 3. Expanding-window walk-forward. The 2018–2026 block is evaluated once, after every design choice is frozen.

The test period is examined once; anything changed afterwards is labelled exploratory.

6. Evaluation

All scores are strictly out of sample and computed on the same origins for every forecast. This report gives them for the development pseudo-out-of-sample period.

Area	Metrics
Statistical accuracy	MSE and QLIKE on the variance scale (QLIKE primary; Patton, 2011); MAE; out-of-sample R^2 against the HAR training-window mean

Area	Metrics
Significance	Diebold–Mariano tests with Newey–West variance and HLN correction, against HAR and against Persistence; Model Confidence Set at 10% over Persistence, HAR, EN and RF (Hansen, Lunde & Nason, 2011)
Attribution	Loss ratios vs HAR and vs Persistence by feature set and horizon; RF-minus-EN change in loss from set A to B, C and D (RQ2); calm vs high-RV regimes; normal vs extreme days; stress episodes
Economic value	Realized utility of a volatility-targeting rule (Bollerslev et al., 2018); stylized variance-swap P&L vs VIX²
Robustness	Rolling window; Risk Lab 15-min and QMLE; ES rv1 / kernel / holiday-drop; Oxford-Man pre-2010 jump features; 1996 VIX-only start; FFD-lag variant; untuned forest

Losses. With forecast $\hat{y}_t$ of realized variance y_t over T days, the two losses are (Patton, 2011):

$$\mathrm{MSE} = \frac{1}{T}\sum_{t=1}^{T}(y_t - \hat{y}_t)^2, \qquad \mathrm{QLIKE} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{y_t}{\hat{y}_t} - \ln\frac{y_t}{\hat{y}_t} - 1\right) \tag{12}$$

QLIKE is robust to noise in the volatility proxy, and it penalizes under-prediction more than over-prediction.

Diebold–Mariano. For a loss differential d_t (benchmark loss minus model loss):

$$\mathrm{DM} = \sqrt{\frac{T + 1 - 2h + h(h-1)/T}{T}}\ \frac{\bar{d}}{\sqrt{\widehat{\mathrm{LRV}}(d)/T}} \tag{13}$$

- The long-run variance is Bartlett, with bandwidth $\max(\lceil 1.5h \rceil, 4(T/100)^{2/9})$ to cover the overlap of multi-day targets.
- The square-root factor in front is the Harvey–Leybourne–Newbold small-sample correction.
- p-values come from Student t_{T-1}.

MCS. The $T_{\max}$ statistic at 10%, with a moving-block bootstrap (block length $\max(h, \lfloor\sqrt{T}\rfloor)$, 5,000 draws).

Regimes.

- *High-RV:* the monthly RV lag exceeds its 2005–2017 80th percentile.
- *Extreme days:* the realized target exceeds Q3 + 1.5·IQR (Rahimikia & Poon, 2026).
- *Stress episodes:* August 2015; the euro crisis (Kılıç, 2026); and the proposal's test episodes (Feb 2018, 2020, 2022, Apr 2025).

Economic value: vol-targeting utility.

$$U = \frac{\mathrm{SR}^2}{\gamma}\left(\sqrt{\frac{\mathrm{RV}}{F}} - \frac{\mathrm{RV}}{2F}\right) \tag{14}$$

with $\mathrm{SR} = 0.4$ and $\gamma = 2$, so the maximum is 4% per year, reached at $F = \mathrm{RV}$.

RQ2. With $L_{s,t}^{m}$ the daily loss of model m on set $s \in \{B, C, D\}$, the DM test is applied to

$$d_{s,t} = \left(L_{A,t}^{\mathrm{RF}} - L_{s,t}^{\mathrm{RF}}\right) - \left(L_{A,t}^{\mathrm{EN}} - L_{s,t}^{\mathrm{EN}}\right), \qquad \bar{d}_s > 0 \ \Rightarrow \text{ the forest gains more} \tag{15}$$

Variance swap. A one-month forecast is compared with the strike $K_t \approx (\mathrm{VIX}_t/100)^2$, which we report in vol-points² as VIX_t^2. Selling one unit of variance pays

$$\Pi_t = K_t - 252 \cdot \mathrm{RV}_{t+1:t+22} \tag{16}$$

- **Timing rule:** sell variance when the annualized forecast is below K_t; buy otherwise.
- **SPY adjustment:** SPY RV omits the overnight move, so SPY forecasts are rescaled by the training ratio of mean squared close-to-close return to mean RV.
- **Payoffs:** the actual close-to-close payoff is reported alongside.

7. Results: Development Period

Every number in this section is pseudo-out-of-sample. Two comparison samples are used:

- **All ten forecasts together** (Persistence, HAR, and EN/RF on sets A–D): **2 January 2015 – 29 December 2017 ($n = 755$)**, for both targets. This short window is forced by the five-year burn-in.
- **SPY sets A–C:** a longer common sample, **2010–2017 ($n = 2{,}013$)**.

Ratios are model loss ÷ HAR loss, so values below 1 beat HAR. Table notation:

- **bold** = significantly lower QLIKE than HAR (one-sided DM, $p < 0.05$);
- ✓ = in the 10% MCS;
- † = significantly lower QLIKE than Persistence.

7.1 Forecast accuracy

Target	h	HAR QLIKE	Persist.	EN A	RF A	EN B	RF B	EN C	RF C	EN D	RF D
SPY	1	0.254	0.952	1.000	**0.815** ✓†	1.675	**0.740** ✓†	1.596	**0.709** ✓†	0.889 ✓	**0.714** ✓†
SPY	5	0.263 ✓	1.078 ✓	1.000 ✓	1.036 ✓	1.335 ✓	**0.919** ✓	1.629 ✓	1.052 ✓	**0.885** ✓	0.928 ✓
SPY	22	0.317 ✓	0.884 ✓	1.000 ✓	0.922 ✓	**0.854** ✓	0.940 ✓	0.955 ✓	1.470 ✓	1.107 ✓	1.078 ✓
ES	1	0.217	0.943	1.021	**0.874** ✓†	**0.800**	**0.630** ✓†	0.861	**0.628** ✓†	**0.759** ✓	**0.633** ✓†
ES	5	0.292	0.895 ✓	1.018	**0.887** ✓	**0.657** ✓†	**0.812** ✓	**0.735** ✓	**0.781** ✓	**0.721** ✓	**0.764** ✓
ES	22	0.325 ✓	0.942 ✓	1.185 ✓	0.967 ✓	0.943 ✓	1.016 ✓	1.084 ✓	1.052 ✓	1.173 ✓	1.064 ✓

- **1 day.** The random forest on any set beats both HAR and Persistence. QLIKE falls 26–29% for SPY (sets B–D) and 37% for ES, and both HAR and Persistence drop out of the MCS.
- **5 days.** Several models beat HAR; for ES, every enriched model does. But only the elastic net on set B for ES also beats Persistence significantly.
- **22 days.**
  - For SPY, the elastic net on set B beats HAR ($p = 0.04$). Nothing beats Persistence, which itself has a lower QLIKE than HAR (0.884).
  - For ES, nothing beats either benchmark.
- **Persistence vs HAR.** On 2015–2017, HAR does not significantly beat naive Persistence at any horizon. Under the proposal's rule, HAR itself is therefore "uninformative" on this window. Section 7.5 shows why: 2015–2017 was mostly calm, and Persistence wins on calm days but loses heavily on volatility spikes. On the longer SPY sample (2010–2017), HAR beats Persistence at every horizon (Persistence ratio 1.06 / 1.09 / 1.10).
- **MSE.** Under MSE, no improvement over HAR is significant anywhere, and every model is in the MSE MCS.
- **Out-of-sample R^2:**

	$h = 1$	$h = 5$	$h = 22$
SPY: HAR	0.33	0.48	0.57
SPY: best model	0.44	0.59	0.69
ES: HAR	0.26	0.26	0.24
ES: best model	0.46	0.43	0.36

Longer SPY sample, sets A–C (2010–2017, $n = 2{,}013$):

h	Persistence	EN A	RF A	EN B	RF B	EN C	RF C
1	1.06	1.00	**0.89**	2.93	**0.80**	2.63	**0.78** ✓
5	1.09 ✓	1.00 ✓	0.99 ✓	2.66	**0.91** ✓	3.16	0.97 ✓

h	Persistence	EN A	RF A	EN B	RF B	EN C	RF C
22	1.10 ✓	1.01 ✓	0.97 ✓	1.91	0.98 ✓	3.98	1.27 ✓

- **1 day:** the forest gains 11–22%.
- **5 days:** the forest on set B gains 9%.
- **22 days:** HAR is not beaten.
- **The elastic net on sets B and C** is far worse than HAR. It hits the forecast floor on 8–12% of days (Section 7.7).

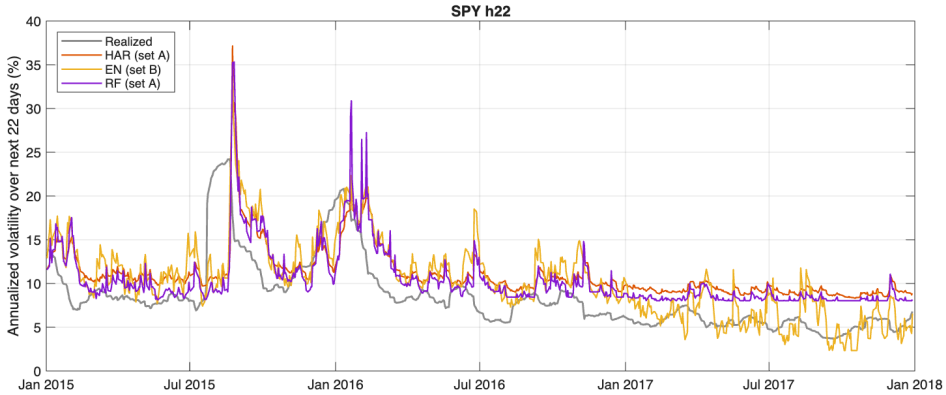


Figure 4. SPY target, $h = 22$: realized 22-day volatility and the HAR, best elastic-net and best random-forest forecasts, 2015–2017.

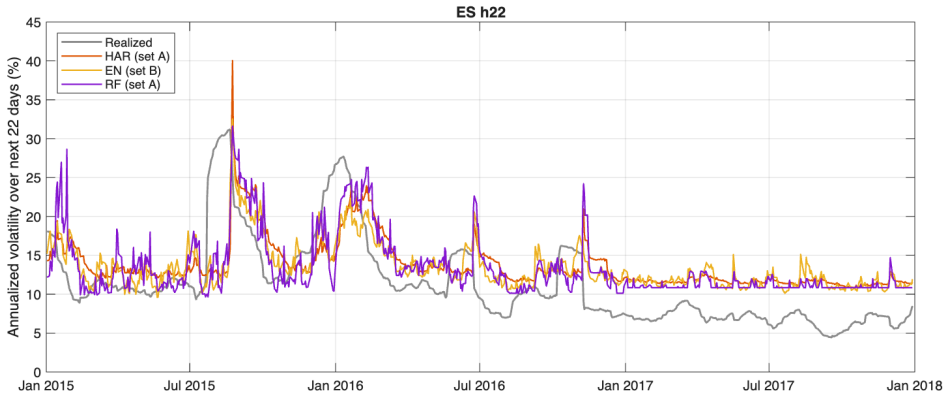


Figure 5. ES target, $h = 22$. In record-low-volatility 2017, every model stays above realized volatility: no model forecasts below the range it was trained on.

7.2 RQ1: volatility history only (set A)

The table gives QLIKE ratios between the four set-A forecasts, with the one-sided DM p-value for "first beats second" in parentheses. SPY uses 2010–2017 ($n = 2{,}013$); ES uses 2015–2017 ($n = 755$).

Comparison	SPY $h = 1$	SPY $h = 5$	SPY $h = 22$	ES $h = 1$	ES $h = 5$	ES $h = 22$
HAR vs Persistence	0.943 (0.14)	0.916 (0.12)	0.906 (0.31)	1.061 (0.78)	1.118 (0.85)	1.062 (0.60)
EN A vs HAR	1.000 (0.91)	1.002 (0.92)	1.010 (0.86)	1.021 (1.00)	1.018 (1.00)	1.185 (1.00)
RF A vs HAR	**0.887** (<0.001)	0.988 (0.39)	0.971 (0.36)	**0.874** (0.008)	**0.887** (0.004)	0.967 (0.24)
RF A vs Persistence	**0.837** (<0.001)	0.905 (0.06)	0.880 (0.17)	**0.927** (0.036)	0.992 (0.46)	1.027 (0.55)

Answer on development data:

- **The elastic net on set A adds nothing.** Validation picks almost no shrinkage, so it reproduces HAR.

- **The random forest beats HAR** at 1 day for both targets and at 5 days for ES. It is the only set-A forecast that significantly beats Persistence (at 1 day).
- **HAR's own advantage over Persistence** is not significant on these windows.
- **The horizon hypothesis is not supported.** Gains are largest at 1 day and do not grow with the horizon.

7.3 RQ2: does the random forest gain more from richer features?

The table gives the QLIKE gain from set A to set s, as a % of each model's set-A loss. It also gives the difference-in-differences (eq. 15) and its one-sided p-value for "RF gains more".

Target (sample)	h	Step	EN gain	RF gain	DiD	p_1
SPY (2010–17)	1	A→B / A→C	−193% / −163%	+10% / +12%	+0.43 / +0.37	< 0.001
SPY (2015–17)	1	A→D	+11%	+13%	−0.00	0.54
SPY (2010–17)	5	A→B / A→C	−165% / −215%	+8% / +2%	+0.35 / +0.44	< 0.001
SPY (2015–17)	5	A→D	+12%	+10%	−0.00	0.56
SPY (2010–17)	22	A→B / A→C	−89% / −294%	−1% / −31%	+0.23 / +0.69	0.015 / 0.022
SPY (2015–17)	22	A→D	−11%	−17%	−0.02	0.60
ES	1	A→B / A→C / A→D	+22% / +16% / +26%	+28% / +28% / +28%	+0.01 / +0.02 / −0.01	0.39 / 0.14 / 0.61
ES	5	A→B / A→C / A→D	+36% / +28% / +29%	+9% / +12% / +14%	−0.08 / −0.05 / −0.05	> 0.99
ES	22	A→B / A→C / A→D	+21% / +9% / +1%	−5% / −9% / −10%	−0.10 / −0.06 / −0.04	> 0.79

Answer on development data. The forest is **more robust** to richer, non-stationary features than the elastic net, but it does **not extract more information** from them:

- **The significant positive DiD values are elastic-net failures.** They occur for SPY on sets B and C, where the elastic net's QLIKE becomes two to four times worse while the forest's barely moves.
- **Where neither model breaks down, the forest never gains significantly more.** This covers set D for SPY and every ES set. For ES at 5 and 22 days the elastic net gains clearly more; for example, from set B at 5 days the elastic net gains +36% and the forest +9%.
- **ES at 1 day:** both models gain substantially (16–28%), the forest slightly more but not significantly.

7.4 RQ3: ES target and the variance-swap check ($h = 22$)

The table gives the mean P&L per 22-day variance-swap trade (vol-points², actual close-to-close payoff) and its t-statistic, for 2015–2017. ρ is the correlation between the forecast signal $K - F$ and the realized payoff $K - \text{realized}$.

Strategy	SPY-target forecasts	ES-target forecasts
Always sell variance	+80.1 ($t = 4.1$)	+80.1 ($t = 4.1$)
Persistence	+42.7 ($t = 2.0$), $\rho = 0.07$	+24.9 ($t = 1.2$), $\rho = 0.04$
HAR	+13.5 ($t = 1.0$), $\rho = 0.09$	+24.4 ($t = 1.7$), $\rho = 0.27$
RF A	+28.5 ($t = 1.6$), $\rho = -0.08$	+39.8 ($t = 3.5$), $\rho = 0.23$
EN B	+23.8 ($t = 1.8$), $\rho = -0.09$	+40.5 ($t = 1.9$), $\rho = 0.42$
RF B	+35.0 ($t = 1.5$), $\rho = -0.09$	+38.9 ($t = 1.8$), $\rho = 0.36$
EN C	+37.4 ($t = 1.9$), $\rho = 0.22$	+27.7 ($t = 1.3$), $\rho = 0.36$
EN D	+25.3 ($t = 1.2$), $\rho = 0.35$	+21.6 ($t = 1.0$), $\rho = 0.31$
RF D	+21.5 ($t = 1.1$), $\rho = 0.37$	+22.4 ($t = 1.1$), $\rho = 0.33$

- **Always selling variance is unbeaten** in calm 2015–2017, which had a large variance risk premium.
- **Some profits are premium harvesting, not timing.** Persistence and the SPY-target forest make money mostly by being short (84–91% and 72–78% of the time). Their signals are uncorrelated, or negatively correlated, with the payoff.
- **ES-target forecasts line up with the swap payoff more consistently.** Every ES model has a positive signal–payoff correlation (0.23–0.42), against −0.09 to 0.37 for SPY. HAR's rises from 0.09 on SPY to 0.27 on ES. This is directionally consistent with RQ3, but it is not yet strong evidence.

7.5 Regimes, extreme days and episodes

QLIKE ratio vs HAR on 2015–2017:

Sample	**SPY** $h = 1$	**SPY** $h = 22$	**ES** $h = 1$	**ES** $h = 22$
Calm (monthly RV ≤ 80th pct)	RF C/D 0.70; Persist. 0.94	Persist. 0.85; EN B 0.84	RF C 0.61; Persist. 0.93	Persist. 0.88; ML 0.97–1.22
High RV (62 SPY / 100 ES days)	EN D 0.68; Persist. 1.15	EN D 0.36, RF D 0.49; Persist. 1.49	RF B 0.79; Persist. 1.08	RF B 0.59, EN D 0.60; Persist. 1.77
Normal days (target ≤ Q3 + 1.5·IQR)	RF C 0.72	**Persist. 0.41**; RF B 0.73	RF C 0.71	**Persist. 0.63**; EN B 0.91
Extreme days (target > Q3 + 1.5·IQR)	EN B 0.45, RF D 0.54	**Persist. 2.47**; RF C 0.80, EN C 0.87	RF B 0.35, RF D 0.37	**Persist. 1.77**; ML 1.03–1.56
August 2015 (54 days)	ML 0.47–0.63	EN C 0.89, RF C 0.71	RF B/C 0.44	RF A 0.62

- **Persistence's strength is a calm-market effect.** At 22 days it beats HAR by more than half on normal days, but it is 1.8–2.5 times worse on extreme days and in high-RV regimes. In mostly calm 2015–2017, the calm days dominate the average.
- **At 1 day, ML is *better* than HAR on extreme days** (0.35–0.54). This is the opposite of Rahimikia & Poon's single-stock finding.
- **At 22 days, most ML models lose to HAR on extreme days,** matching their finding.

7.6 Economic value: volatility-targeting utility

Realized utility (eq. 14) in % per year; the maximum is 4.00.

Target	h	**Persistence**	**HAR**	**Best model**
SPY	1	3.46	3.51	3.66 (RF D)
SPY	5	3.33	3.47	3.52 (EN D)
SPY	22	3.33	3.42	3.46 (EN B)
ES	1	3.52	3.56	3.74 (RF B, C, D)
ES	5	3.39	3.41	3.59 (EN B)
ES	22	3.30	3.38	3.41 (EN B)

- **HAR ranks above Persistence at every horizon,** because utility penalizes the large spike errors Persistence makes.
- **The best ML model ranks above both.** Its advantage is about 0.15–0.18 percentage points per year at 1 day, 0.05–0.18 at 5 days, and 0.03–0.04 at 22 days.

7.7 Forecast floors and fractional differencing in practice

- **Floors.** The elastic net on sets B and C hits the training-minimum floor on 9–12% of SPY days at 1 day (2015–2017), and on 8–11% over 2010–2017. HAR, the elastic net on sets A and D, the forest and every ES model hit it on 2% of days or fewer. QLIKE punishes under-prediction heavily, so floored days account for most of the elastic net's losses on sets B and C.
- **FFD inside the walk-forward:**
  - differenced in every fold: the 10Y yield, the 10Y−3M spread and ATM 1Y implied vol (d^* 0.10–0.25);
  - differenced in two-thirds of folds: ATM 3M implied vol;

- differenced occasionally: CISS, ATM 1M IV, log RVX and STLFSI4.

Retained correlation is 0.96–0.999. FFD does not cure the elastic net's instability on set B, because that comes from the *stationary* but collinear implied-vol levels (Section 8).

7.8 What drives the forecasts

- **Which part of set B does the work** (the proposal's implied-vol question).
  - *Forest, SPY, 1 day:* the IV term slope (27%), the weekly negative-return average (13%) and daily RV (13%).
  - *Forest, ES, 1 day:* the VIX9D−VIX spread (23%) and daily RV (20%).
  - *Forest, 22 days:* the IV term slope, skew and cross-market implied vols lead.
  - *Elastic net:* at 1 day, its set-B coefficients load on the Parkinson range and the term slope. At 22 days they are large and offsetting on ATM implied vol (1M +0.48, 3M −0.31), the signature of collinearity.
  - **Conclusion: implied volatility does help, mainly through its shape (term slope, short-dated spread, skew) rather than its level.**
- **Set D, random forest:**
  - SPY, 1 day: ES continuous variation (9%), IV skew (7%), weekly negative returns (7%), VIX9D spread (7%).
  - ES, 1 day: VIX9D spread (23%), ES negative semivariance (10%), realized kernel (8%).
- **Set D, elastic net** (selected in every fold):
  - 1 day: the Parkinson range, VIX9D spread, ES negative semivariance, VRP, realized kernel and (negatively) the daily negative return.
  - 22 days: news sentiment (negative), IV skew, the monthly RV lag and the term slope.
- **Macro and calendar features rank at the bottom** for both models (ADS, EPU, GPR, Hang Seng, calendar flags). This echoes Kılıç (2026).
- **Shapley values** for the last-fold forest (2017, $h = 22$, set D) spread importance thinly:
  - SPY: NFCI, the weekly RV lag and the Parkinson range;
  - ES: ATM 3M and 1Y implied vol, the Parkinson range and CISS.
- **ALE curves** (Figure 6) are steep at low values and flat above: the forest cannot extrapolate.

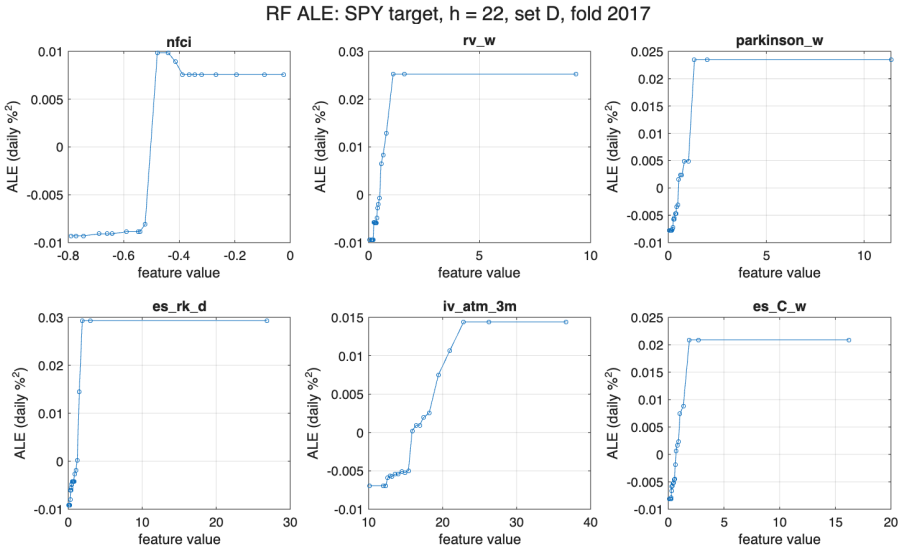


Figure 6. Accumulated local effects of the six most important features, random forest on set D, SPY target, $h = 22$ (2017 fold).

7.9 Memory in forecast errors

The table gives the local Whittle $\hat{d}$ ($m = T^{0.65}$, s.e. 0.05) and the Ljung–Box $Q(10)$ p-value for the 1-day forecast errors, 2015–2017.

Target	Forecast	$\hat{d}$ of errors	Ljung–Box p

Target	Forecast	$\hat{d}$ of errors	Ljung–Box p
SPY	Persistence / HAR	−0.49 / −0.33	< 0.001 / 0.001
SPY	RF A–D	−0.08 to −0.37	≤ 0.014
SPY	EN B, C, D	−0.06 to 0.03	0.56, 0.31, 0.25 (white)
ES	Persistence / HAR	−0.49 / 0.02	< 0.001
ES	EN B–D, RF A–D	−0.15 to 0.10	≤ 0.003 (EN D: 0.15)

- **No forecast leaves long memory in its errors.** Every $\hat{d} \leq 0.10$, and the autocorrelation tails are noise: tail $R^2 < 0.06$, against Bhatti's 0.6 threshold.
- **Short-range dependence remains,** as a negative first-order autocorrelation (Figure 2, right). After a volatility spike, the next forecast is too high. This is the measurement-error effect HARQ was designed for (Bollerslev, Patton & Quaedvlieg, 2016), and that set D's $\sqrt{\mathrm{RQ}} \times \mathrm{RV}$ targets.
- **Persistence errors are strongly anti-persistent** ($\hat{d} \approx -0.5$), as expected when forecasting with a single noisy day.

7.10 Robustness

Each check reruns the walk-forward with one change. The table gives QLIKE ratios against that scenario's own HAR, on 2015–2017 unless stated. **Bold** = significantly better than HAR; ✓ = in the MCS; † = significantly better than Persistence.

Check (scenario)	$h = 1$	$h = 5$	$h = 22$
Rolling window, SPY (10 years)	RF C **0.68** ✓†, RF D **0.69** ✓†	EN D **0.84** ✓, RF D **0.88** ✓	best RF A 0.86 ✓; Persistence 0.81
Rolling window, ES (8 years)	RF C/D **0.63** ✓†	EN D **0.72** ✓, EN C **0.73** ✓	none below 0.97
SPY target = Risk Lab 15-min RV	RF D **0.70** ✓†, RF C **0.73** †	EN D / RF D **0.88** ✓	EN C **0.87** ✓; Persistence 0.77
SPY target = Risk Lab QMLE	RF C/D **0.70** ✓†, EN D **0.75** ✓†	EN D **0.84** ✓, RF D **0.85** ✓	best 0.87 (n.s.); Persistence 0.76
ES target = 1-min RV	RF D **0.66** ✓†, RF C **0.72** ✓†	EN C/D **0.72** ✓	none below 0.98
ES target = realized kernel	RF D **0.70** ✓†, RF C **0.71** ✓†	EN D **0.72** ✓, RF D **0.75** ✓	none below 0.97
ES holiday rows dropped	RF D **0.62** ✓†, RF C **0.64** ✓†	EN D **0.72** ✓, EN C **0.73** ✓	best RF A 0.92 (n.s.)
Random forest with untuned defaults	SPY RF D **0.68** ✓†; ES RF D **0.60** ✓†	SPY RF D 0.95 ✓; ES RF D **0.76** ✓	ES RF A **0.84** ✓; sets C/D 1.31–1.46
FFD-lag variant (ML lag block = FFD log RV)	SPY RF C **0.68** ✓†; ES RF B/C **0.64** ✓†; EN on FFD lags alone 1.34–2.08	SPY RF B **0.86** ✓†; ES EN B **0.68** ✓	ES EN B **0.91** ✓; otherwise ≥ 0.94
Oxford-Man jump features, SPY 2010–2017 ($n = 2{,}013$)	RF C+OM **0.77** ✓† vs RF C **0.78** ✓†	RF C+OM **0.90** ✓† vs RF C 0.97	1.21 vs 1.27
1996 start, HAR lags + log VIX (2001–2017, $n = 4{,}276$)	RF A **0.84** ✓†, RF A+VIX **0.84** ✓†	RF A+VIX **0.81** ✓; Persistence **0.83** ✓	EN A+VIX **0.82** ✓, RF A+VIX **0.87** ✓; Persistence 0.86

What the checks show:

- **The 1-day result is robust.** In every scenario the best forest beats both HAR and Persistence, at ratios of 0.60–0.84. This holds across alternative targets (15-min RV, QMLE, 1-min RV, realized kernel), windows, holiday treatment, untuned forests and FFD lags.
- **At 5 days, set-D models beat HAR in every scenario that runs them** (the untuned SPY forest, 0.95, is the one exception). Only the SPY FFD-lag and Oxford-Man variants also beat Persistence.
- **At 22 days no robustness check produces a forecast that beats Persistence.** Persistence itself often has lower QLIKE than HAR (0.76–0.94).

- **Persistence vs HAR over the long 2001–2017 window** (1996-start check): Persistence significantly beats HAR at 5 days (0.83). HAR on variance levels, fitted by least squares, is a weak QLIKE forecaster at medium horizons.
- **FFD lags.** The forest is indifferent to whether its lag block is raw RV or FFD log RV. The elastic net is not: on FFD log RV alone it is much worse than HAR, because a linear model in a log-scale input cannot reproduce a level-scale target. Keeping the raw HAR lags, as the proposal specifies, matters for the linear model.
- **Smaller checks.** Untuned forests match tuned ones at 1 and 5 days but are worse at 22 days on the large sets. Oxford-Man jump features add a little at 5 days for SPY. Dropping ES holiday rows changes nothing material.

7.11 Success-criterion scorecard (development data)

Target	h	Best ML	Ratio vs HAR	p vs HAR	In MCS	Beats HAR	Ratio vs Persist.	Beats Persist.	Swap timing (P&L, ρ)
SPY	1	RF C	0.709	< 0.001	yes	**yes**	0.745	**yes**	—
SPY	5	EN D	0.885	0.003	yes	**yes**	0.821	no	—
SPY	22	EN B	0.854	0.036	yes	**yes**	0.966	no	+23.8, $\rho = -0.09$ (not consistent)
ES	1	RF C	0.628	< 0.001	yes	**yes**	0.666	**yes**	—
ES	5	EN B	0.657	< 0.001	yes	**yes**	0.734	**yes**	—
ES	22	EN B	0.943	0.10	yes	no	1.001	no	+40.5, $\rho = 0.42$ (consistent)

- **On development data the success criterion is met.** ML beats HAR significantly, with the winner in the MCS, at five of six target–horizon pairs, and the ES variance-swap check is directionally consistent.
- **Against the Persistence floor, far less is informative.** Only the 1-day forests (both targets) and the 5-day elastic net on ES set B pass. At 22 days, and at 5 days for SPY, no forecast (HAR included) significantly beats naive Persistence on 2015–2017.
- **The locked test will be far more decisive** than this calm three-year window: it covers nine years and the 2018, 2020, 2022 and 2025 stress episodes.

8. Discussion and Relation to the Literature

The project folder holds the three reference papers and the long-memory working paper cited in Section 4.3.

Study	Data and design	Headline finding	What this project takes from it
Christensen, Siggaard & Veliyev (2023), *JFEC* (arXiv:2601.13014)	29 DJIA stocks, 5-min RV, 2001–2017; averaged targets at 1, 5, 22 days; HAR family vs EN, RF, boosting, neural nets; MSE, DM, MCS 90%	With HAR lags only ML ≈ HAR; with covariates RF MSE 0.90 / 0.80 / 0.60 of HAR; gains grow with the horizon	Floor rule; train-only standardization; trailing validation (no shuffling); RF-defaults check; ALE
Kılıç (2026), FEDS 2025-061r1	S&P 500 RV 1996–2025 and 40 stocks; HAR, ARFIMA, regime-switching HAR vs XGBoost, MLP, LSTM; QLIKE, DM–HLN, MCS	ML improves on HAR by about 1% at most at 1 day and loses by 22 days; extended predictors do not help	Label-availability rule; HLN-corrected DM; MCS block length $\lfloor\sqrt{T}\rfloor$; euro-crisis window; utility check; EN coefficients as a result
Rahimikia & Poon (2026), SSRN 3707796	23 NASDAQ stocks with order-book and news data, 2007–2022, 1 day; HAR family (CHAR) vs LSTM	ML loses overall, wins on normal days, loses badly on extreme days (RV > Q3 + 1.5·IQR)	Extreme-day split; floor-frequency reporting
Bhatti (2026), Fordham working paper	Long memory in S&P 500 constituent log prices; finite-sample detectability boundary	$d_0(T) = 5.81T^{-0.367}$; ADF passes after one difference while KPSS keeps rejecting under long memory	ADF and KPSS read together; d_{LTM} cross-check with its caveats (Section 4.3)

Against Christensen et al. They find large 22-day gains under MSE; we find none that survive the Persistence floor under QLIKE. Our set-D MSE ratios at 22 days for SPY (0.71–0.72) are close to theirs, but not significant over three years. Two design differences matter: they forecast individual stocks and refit daily, while we forecast an index and refit annually.

With Kılıç. As he finds, ML gains are concentrated at short horizons, and extended macro predictors add little.

Against Rahimikia & Poon. At 1 day our models do *better* than HAR on extreme days. Index RV is smoother than single-stock RV, and our models see the IV term structure and ES semivariances. At 22 days, though, the extreme-day loss reappears.

Why the elastic net breaks down on sets B and C.

1. Validation sometimes picks an almost-zero penalty.
2. The near-OLS fit then puts large, offsetting coefficients on the collinear implied-vol block (VIX, ATM 1M/3M/1Y IV, VRP).
3. These inputs mostly pass the ADF test, so FFD leaves them unchanged.
4. When they move one to two standard deviations away from their training range, the linear forecast falls below zero and is floored.

Set D is unaffected because the ES realized kernel carries most of the signal. Possible remedies would change the specification and must be decided before the test:

- a floor on λ;
- an unpenalized HAR block, as in Kılıç;
- trimming the implied-vol block.

Why the forest cannot forecast outside its range. A tree forecasts the average of the training targets in a leaf, which bounds forecasts by the training range. This is visible in 2017 (Figure 5). An earlier configuration with a three-year burn-in (proposal v5.2) exposed the reverse case. Trained only on 2005–2007, the forest's QLIKE in the 2008–2009 crisis was about six times HAR's. The five-year burn-in of v5.4 starts development forecasts in 2010, after the crisis has entered training.

Why Persistence is strong on 2015–2017. In a persistently calm regime, the latest realized variance is an excellent forecast, while HAR's monthly lag and intercept pull forecasts toward the higher training average. Persistence loses badly whenever volatility jumps (Section 7.5). Utility punishes those errors, which is why it ranks HAR above Persistence.

9. Applications

- **Volatility and variance trading.** A forecast of realized variance against VIX^2 indicates when implied volatility is rich or cheap. The ES results suggest full-session targets suit this better.
- **Volatility targeting and risk budgeting.** The utility results quantify what better forecasts are worth to a fund that scales exposure by $1/\hat{\sigma}$: about 0.15 percentage points per year at the 1-day horizon.
- **Risk limits and margining.** One-day forecasts feed value-at-risk limits and initial-margin models. That is the horizon where the ML gains are largest and clear both benchmarks.
- **A reusable evaluation harness.** Any new volatility model can be dropped into `walk_forward.m` and judged against HAR and Persistence under the same rules: label gap, train-only stationarity transforms, tuning protocol, DM and MCS tests, regimes and economic checks.

10. Risks, Limitations and Open Decisions

Risk (proposal Section 7)	Mitigation in the implementation	Status
Look-ahead bias	Publication-lag table; real-time ADS; label gap; train-only standardization, imputation, floors and d^*; locked test	In place
Overfitting a flexible model	Nested sets; DM and MCS; design frozen before the test; exploratory label for later ideas	In place
ES session and sample	SPY remains the main target; Oxford-Man pre-2010 check; holiday-row robustness	In place
Over-differencing or look-ahead in choosing d	Smallest d passing ADF, re-chosen inside each training window; retained correlation reported (0.96–0.999); raw HAR lags kept; FFD-lag variant as robustness	In place
Missing SPY days in test	Targets never imputed; counts measured (Section 3.2); ES models scored on those days	Treatment to be decided

Risk (proposal Section 7)	Mitigation in the implementation	Status
Overnight mismatch	ES secondary target for the swap payoff; SPY forecasts rescaled in that check only	In place
Scope for one semester	Three models plus two unestimated benchmarks	In place

Limitations.

1. **Development results only.** The locked test has not been run.
2. **A short, calm comparison window.** The all-model window is 2015–2017: 755 days, with only 60–100 high-RV days.
3. **Levels-scale modelling** causes both the forest's extrapolation problem and the elastic net's floor problem.
4. **A narrow benchmark set.** The benchmarks are plain HAR and Persistence. Richer econometric models (ARFIMA, CHAR, regime-switching HAR) might erase the ML gains, so our conclusions are "relative to HAR and Persistence".
5. **Tuning noise.** Twelve Bayesian-optimization steps with one seed make the forest's hyperparameters noisy across years.
6. **Data gaps.** There are no limit-order-book or news-text data; ES session times are inferred; and pre-December-2008 ADS is the revised vintage (training rows only).
7. **A stylized variance-swap check.** VIX^2 only approximates the strike; there are no costs; trades overlap.
8. **Many tests.** Read individual p-values together with the MCS.
9. **Hand-coded tests.** ADF and KPSS are coded directly rather than taken from the Econometrics Toolbox. They reproduce the textbook behaviour on simulated white noise and random walks.
10. **An early look at two benchmarks on the test period.** An earlier version of Aryan's independent HAR script also scored HAR and Persistence on 2018–2026 (20 October 2026). Neither benchmark has tuning parameters, and no design choice in this report uses those numbers. The integrated script now follows the same test lock as the main pipeline, and its committed outputs cover development years only. We disclose this so the locked test can be read with it in mind.

Decisions to take before unlocking the test.

1. **Missing-day treatment** (proposal 3.2, explicitly open). Confirm or replace the provisional rules:
   - the 20% rule for multi-day targets;
   - the 80% rule for weekly and monthly lags;
   - dropping origins with a missing RV_t (33 test origins at $h = 1$; 531 have a missing day somewhere in the 22-day lag window);
   - last-published values for inputs on market holidays.
2. **ES holiday sessions** (proposal 3.2): obtain the professor's confirmation of rolling them into the next NYSE day.
3. **Elastic-net stability:** keep the proposal's specification, or add a guard (Section 8).
4. **Random-forest tuning budget:** more `bayesopt` steps, or averaged seeds.
5. **Optional boosting run** on the winning feature set.
6. **Freeze and run:** tag the code version, set `cfg.runLockedTest = true`, run `./run_parallel.sh test` once, and archive the outputs.

11. Implementation Plan and Code

Work proceeded in thin slices, so that each phase left a runnable pipeline:

1. Persistence and HAR on set A, end to end;
2. the elastic net and random forest;
3. richer feature sets.

All code is in `Code/` (MATLAB R2026a with the Statistics and Machine Learning Toolbox; neither the Parallel Computing nor the Econometrics Toolbox is needed).

Task (proposal Section 8)	Implementation

Task (proposal Section 8)	Implementation
Targets, Persistence and HAR lags	`load_panel.m` (NYSE-calendar alignment), `build_design.m` (targets, features, publication lags), `fit_persistence.m`, `fit_har.m`
Stationarity and FFD	`adf_test.m`, `kpss_test.m`, `ffd.m`, `ffd_choose.m` (coded directly); `stationarity.m` (diagnostics); d^* per fold in `walk_forward.m`
Elastic net	`fit_en.m` (`lasso` with Alpha, plus closed-form ridge)
Random forest	`fit_rf.m` (`TreeBagger`, `bayesopt`)
Walk-forward and jobs	`walk_forward.m`, `run_job.m`, `make_spec.m` (main and robustness scenarios), `rv_config.m` (every setting and the test lock)
Evaluation and explanation	`evaluate_phase.m`, `qlike.m`, `dm_test.m`, `mcs.m`; `explain_rf.m` (Shapley, ALE); `long_memory.m`; `data_audit.m`
Drivers and report	`run_parallel.sh`, `main.m`; `report_figures.m`, `Report/build_report.py`
Team modules and integration check	`Code/Robin/` (Robin: targets, aligned features, feature sets A–D as standalone files); `aryan/code/rv_har_pipeline.m` (Aryan: independent HAR vs Persistence with its own rules); `Code/check_integration.m`

Integration check (21 October 2026). Robin's standalone data preparation reproduces the main pipeline's 46 features and three targets to within 10^{-14} for both targets. Aryan's independent HAR and Persistence forecasts have correlations of 0.999–1.000 with the main pipeline's on development data, and their HAR/Persistence QLIKE ratios differ by at most 0.01, even though his script uses its own missing-day, holiday and floor rules.

How to run.

- `./run_parallel.sh dev` regenerates the main development results (about 15 minutes on a 14-core Mac).
- `./run_parallel.sh dev robustness` runs all ten robustness scenarios (about 45 minutes).
- `python3 Report/build_report.py` rebuilds this document.
- The locked test runs only after `cfg.runLockedTest = true` is set.

Implementation choices the proposal leaves open (all in `rv_config.m`, `build_design.m` or `walk_forward.m`):

Choice	Setting
Units	Variances in daily %²
Tuning criterion	Validation QLIKE
FFD grid and threshold	$\{0, 0.05, \ldots, 1\}$; $\tau = 10^{-3}$
ADF specification	Constant; Schwert lag length
Late-starting inputs	Enter a fold once they have three years of history; earlier rows get the training median
Calendar flags	Next-session early close; days to the next session
DM bandwidth	$\max(\lceil 1.5h \rceil, 4(T/100)^{2/9})$
MCS bootstrap	Block length $\max(h, \lfloor\sqrt{T}\rfloor)$; 5,000 draws
High-RV threshold	80th percentile of the monthly RV lag
SPY rescaling in the swap check	Training ratio of close-to-close variance to RV

Team member	Lead responsibility	Focus
Robin	Risk Lab and VOLARE ES targets, feature alignment and publication lags, data dictionary, QC	Data
James	Persistence and HAR; walk-forward engine; DM tests and Model Confidence Set	RQ1 benchmarks

Team member	Lead responsibility	Focus
Aryan	Elastic net and random forest, tuning, coefficients and feature importance	RQ1–RQ2 ML
All	Volatility-timing utility, variance-swap comparison, report and presentation	RQ3 / write-up

Calendar.

Milestone	Status
Data lock (mid-October)	Done
Persistence, HAR and first ML baseline (end-October)	Done
Full sets A–D and ES runs (mid-November)	Done on development data
Locked test (late November)	Pending the decisions above
Economic checks, final report and slides (through early December)	Report drafted on development data

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Data sources: Dacheng Xiu, Risk Lab (research use); VOLARE (volare.unime.it), ES measures downloaded 4 October 2026 (research use); Oxford-Man Institute Realized Library (archived mirror); Cboe; FRED; Federal Reserve Bank of Philadelphia (ADS); ECB (CISS); SF Fed news sentiment; Caldara–Iacoviello GPR; Yahoo Finance; Bloomberg.

Appendix A. Feature Dictionary

All variances are in daily %². "Lag" is the publication shift in trading days. "FFD" says whether the input is a fractional-differencing candidate; every input is one except the raw HAR lags and the calendar flags. On development data, only the unit-root inputs marked † are actually differenced in most folds.

Feature	Set	Definition	Source	Lag	FFD
`rv_d`, `rv_w`, `rv_m`	A	Target RV today; 5-day and 22-day averages (≥80% observed)	Risk Lab `rv5_trade_vol`²/252 (SPY) or VOLARE `rv5` (ES)	0	never (raw)
`negret_d`, `negret_w`, `negret_m`	B	$\min(r_t, 0)$ of the S&P 500 close-to-close % return; 5- and 22-day averages	`spx_logret_cc_pct`	0	candidate
`parkinson_d`, `parkinson_w`	B	Parkinson range variance; 5-day average	`spx_parkinson_var_daily`	0	candidate
`log_vix`, `vrp`	B	$\ln\mathrm{VIX}$; $\mathrm{VIX}^2/252 - \mathrm{RV}^{(m)}$	Cboe	0	candidate
`iv_atm_1m`, `iv_atm_3m`, `iv_atm_1y`	B	At-the-money implied vol, vol points	Bloomberg	0	3M†, 1Y†
`iv_skew_1m`, `iv_term`	B	IV(90%) − IV(110%) at 1 month; ATM IV(1Y) − IV(1M)	Bloomberg	0	candidate
`t10y3m`, `dgs10`	B	10Y − 3M Treasury spread; 10Y yield	FRED	0	both†
`vvix`, `skew`, `vix9d_spread`	B	VVIX; Cboe SKEW; VIX9D − VIX	Cboe	0	candidate
`log_vxn`, `log_rvx`	B	Log Nasdaq-100 and Russell 2000 implied vol	Cboe / FRED	0	candidate
`ads`	C	ADS business conditions, real-time vintage	Philadelphia Fed	built in	candidate
`log_epu`	C	Log daily Economic Policy Uncertainty	Baker–Bloom–Davis	1	candidate
`ciss`	C	ECB New CISS for the US	ECB	2	† (some folds)
`news_sent`	C	SF Fed daily news sentiment	SF Fed	5	candidate
`log_gpr`	C	Log classic daily geopolitical risk	Caldara–Iacoviello	5	candidate
`nfci`, `stlfsi4`	C	Chicago Fed NFCI; St. Louis Fed stress index	FRED	built in	candidate
`hsi_sq`	C	Squared Hang Seng return (0 when HKEX is closed)	Yahoo	0	candidate
`early_close_next`, `gap_days_next`	C	Next NYSE session is an early close; calendar days to the next session	NYSE calendar	0	never
`es_C_d/w/m`, `es_J_d/w/m`	D	ES continuous and jump components (eq. 3); 5- and 22-day averages	VOLARE `rv5`, `bv5`	0	candidate

Feature	Set	Definition	Source	Lag	FFD
es_rsp_d/w, es_rsn_d/w	D	ES positive and negative semivariance; 5-day averages	VOLARE rsp5, rsn5	0	candidate
es_sqrtrq_rv	D	$\sqrt{\text{RQ}} \times \text{RV}$ (Bollerslev, Patton & Quaedvlieg, 2016)	VOLARE rq5, rv5	0	candidate
es_rk_d	D	ES realized kernel	VOLARE rk	0	candidate
overnight_d, overnight_w	D	$\max(\text{ES rv5} - \text{SPY RV}, 0)$; 5-day average	VOLARE, Risk Lab	0	candidate
log_rv	L	Log RV of the target (FFD-lag robustness variant only)	as set A	0	candidate

Appendix B. Compliance with Proposal v5.4

Status key:

- **Done:** implemented and run on development data.
- **Pending:** implemented, but waiting for the locked test or for a decision the proposal leaves open.
- **Optional:** allowed by the proposal, not done.

Proposal item (section)	Status	Where
Three models plus Persistence and VIX² benchmarks (Abstract, §1, §4.4)	Done	`fit_persistence.m`, `fit_har.m`, `fit_en.m`, `fit_rf.m`, `evaluate_phase.m`
Four nested sets; HAR and Persistence on A; ML on A–D (§4.2, §4.4)	Done	`build_design.m`, `set_columns.m`, `run_job.m`
Horizons 1, 5, 22; target = average RV over $t+1:t+h$ (§4.1)	Done	`build_design.m`
Main target SPY rv5 from 2005; ES second target and set D from 2010 (§3.1)	Done	`sample_start.m`
NYSE calendar alignment; weekly series held to next release (§3.2)	Done (calendar from `features_daily.csv`; the named calendar file is not in the data folder)	`load_panel.m`
Late starts from first date + ℓ_τ; common days only (§3.2)	Done	`ffd.m`, `walk_forward.m`, `evaluate_phase.m`
Missing-day counts measured (§3.2)	Done (reproduced exactly)	`data_audit.m`
Missing-day treatment (§3.2: "will be decided")	Pending (provisional rules in place)	`rv_config.m`, `build_design.m`
ES holiday rows rolled forward; dropping as robustness (§3.2)	Done / Pending professor's confirmation	`load_panel.m`; scenario `holdrop`
Linear forecasts floored at the training-window minimum RV (§4.1)	Done	`fit_har.m`, `fit_en.m`
ADF and KPSS on every target and input, levels and logs, development sample only (§4.3)	Done (coded directly; Econometrics Toolbox not installed)	`adf_test.m`, `kpss_test.m`, `stationarity.m`
Local Whittle and Bhatti's detectability-boundary estimator for RV memory (§4.3)	Done	`long_memory.m`
FFD with the smallest d passing ADF, chosen inside each training window; τ and grid fixed in advance; d^* and retained correlation reported (§4.3)	Done	`ffd.m`, `ffd_choose.m`, `walk_forward.m`, `ffd_*.csv`
Target never transformed; raw HAR lags kept in both ML models (§4.3)	Done	`walk_forward.m`
FFD variant replacing the ML lag block with FFD log RV (§4.3)	Done	scenario `ffdlags`
Elastic net: α and λ tuned; ridge and lasso special cases; coefficients reported (§4.6)	Done	`fit_en.m`, `explain_*.csv`
Random forest; impurity importance (§4.7)	Done	`fit_rf.m`
Excluded models not estimated (§4.8)	Done	—
Optional exploratory boosting run (§4.8)	Optional	—

Proposal item (section)	Status	Where
Boundaries; re-fit after each year-end; development years once the window holds five years (§5)	Done	`walk_forward.m`, `rv_config.m`
Tuning on the last two years; label gap (eq. 10) (§5)	Done	`walk_forward.m`
Availability timing and publication shifts (§5)	Done	`build_design.m`
Rolling 10-year (8-year ES) robustness (§5)	Done	scenario `rolling`
Locked 2018–2026 test examined once (§5)	Pending (early benchmark look disclosed, Section 10)	`cfg.runLockedTest = false`
MSE, QLIKE, MAE, OOS R^2 on the same origins (§6)	Done	`loss_*.csv`
DM (Newey–West + HLN) against HAR and Persistence; MCS at 10% over Persistence, HAR, EN, RF (§6)	Done	`dm_test.m`, `mcs.m`, `loss_*.csv`, `rq1_*.csv`
Loss ratios vs HAR and vs Persistence; RQ2 difference-in-differences (§6)	Done	`loss_ratios_vs_HAR.csv`, `rq2_*.csv`
Calm vs high-RV regimes; stress episodes (§6)	Done / Pending (test episodes run with the test)	`regimes_*.csv`
Vol-targeting utility; variance-swap P&L vs VIX² (§6)	Done	`utility_*.csv`, `varswap_*_h22.csv`
Robustness: rolling; Risk Lab 15-min and QMLE; ES rv1, kernel, holiday drop; Oxford-Man; 1996 VIX-only (§6)	Done	scenarios `rolling`, `rv15`, `qmle`, `rv1`, `rk`, `holdrop`, `oxford`, `vix1996`
Success criterion, with Persistence as the floor (§6)	Done	`success_criterion.csv`, Section 7.11
Shapley and ALE (§8)	Done	`explain_rf.m`
Code regenerates every table from the cleaned inputs (§8)	Done	`run_parallel.sh`
Final report (§8)	Done (development)	this report
Slides (§8)	Optional (after the test)	—
Additions from the research papers	Done	Extreme-day split (Rahimikia & Poon); RF-defaults robustness and floor reporting (Christensen et al.); MCS block length and euro-crisis window (Kılıç)

Appendix C. Output Files

Location	Contents
`Output/panel.mat`	The aligned daily panel (all inputs on the NYSE calendar)
`Output/forecasts/dev/<scenario>/<target>_h<h>.mat`	Every forecast, with per-fold hyperparameters, kept features, coefficients, importances and FFD orders
`Output/tables/dev/<scenario>/summary.md`	All tables for a scenario in one readable file
`.../loss_*.csv`, `loss_noD_*.csv`	Accuracy, DM against HAR and Persistence, MCS; sets A–C on the longer SPY sample
`.../rq1_*.csv`, `rq2_*.csv`	RQ1 pairwise tests (Persistence, HAR, EN, RF); RQ2 difference-in-differences
`.../regimes_*.csv`, `utility_*.csv`, `varswap_*_h22.csv`	Regimes and episodes; vol-targeting utility; variance swap

Location	Contents
`.../explain_*.csv`, `shap_*.csv`, `ale_*.png`, `ffd_*.csv`	EN coefficients and RF importance; Shapley values; ALE plots; FFD orders per input
`.../stationarity.csv`, `long_memory_*.csv`, `long_memory_acf.png`	ADF/KPSS diagnostics with development d^*; long-memory estimates
`.../success_criterion.csv`, `loss_ratios_vs_HAR.csv`	Success scorecard (with Persistence floor); every loss ratio in one table
`Output/tables/audit_*.csv`	Missing-data counts and the test-origin table of proposal 3.2
`Output/logs/`	One log per MATLAB job

Appendix D. Glossary

Term	Meaning
RV	Realized variance: the sum of squared intraday returns (eq. 1)
Persistence	Naive benchmark: the latest h-day average of RV is the forecast (eq. 7)
HAR	Heterogeneous autoregressive model: regression of future RV on daily, weekly and monthly RV averages (eq. 8)
ADF / KPSS	Unit-root test (null: unit root) / stationarity test (null: stationary)
FFD, d^*	Fixed-width fractional differencing (eq. 5); d^* is the smallest order that makes an input pass ADF
Long memory, d	Very slow (hyperbolic) decay of autocorrelation; d is the fractional-integration order
Local Whittle; d_{LTM}	Semiparametric estimator of d (eq. 6); Bhatti's detectability-boundary estimator
QLIKE	Quasi-likelihood loss (eq. 12); robust to noise in the volatility proxy
DM test	Diebold–Mariano test of equal forecast accuracy, with the HLN correction (eq. 13)
MCS	Model Confidence Set: the models that cannot be distinguished from the best
DiD	Difference-in-differences of losses for RQ2 (eq. 15)
Uninformative	A forecast that does not significantly beat Persistence at that horizon
VRP	Variance risk premium: implied minus expected realized variance
Label gap	A training row is used only once its whole target window has been observed (eq. 11)
Walk-forward	Re-estimate after each year-end on past data only; forecast the next year
Floor	Linear forecasts below the training minimum of daily RV are set to that minimum
ALE / Shapley	Model-agnostic explanations of how features move forecasts